

Professional Appointment

- 2025-present **University of Ottawa**, *Tenure-track Assistant Professor*.
School of Engineering Design and Teaching Innovation
School of Electrical Engineering and Computer Science
Faculty of Engineering
- 2025-present **Vector Institute**, *Faculty Affiliate*.

Education

- 2020-2025 **University of Waterloo**, *Ph.D. in Computer Science*.
Waterloo AI Lab and Vector Institute, Supported by David R. Cheriton Graduate Scholarship
Supervisor: Prof. Yaoliang Yu and Dr. Sun Sun
Thesis: Trustworthy Machine Learning with Data in the Wild
- 2018-2020 **University of Manitoba**, *M.Sc. in Computer Science*.
Computer Vision Lab, Supported by University of Manitoba Graduate Funding
Supervisor: Prof. Yang Wang
Thesis: Anomaly Detection in Surveillance Videos using Deep Learning
- 2013-2017 **University of Electronic Science and Technology of China**, *B.E. in Electronic Engineering*.
Statistical Machine Intelligence and Learning Lab
Supervisor: Prof. Zenglin Xu and Prof. Zhao Kang
- 2016 **University of California, Santa Barbara**, *Electrical Computer Engineering*.
Exchange Program, Funded by China Scholarship Council

Grants

- 2026-2031 NSERC Discovery Grant
- 2026-2027 NSERC Discovery Grant - Launch Supplement
- 2025-2028 University of Ottawa Startup Grant
- 2025-2026 University of Ottawa SEDTI Startup Grant

Award and Honors

- 2024 Apple Scholars PhD fellowship in AI/ML Nomination (1 of 3), University of Waterloo
- 2024 Google PhD Fellowship Nomination (1 of 4), University of Waterloo
- 2024 IEEE Computer Society Honorarium for SaTML (with grant)
- 2024-2025 David R. Cheriton Graduate Scholarship, University of Waterloo
- 2023 NeurIPS 2023 Top Reviewer (with grant)
- 2023 Graduate Student Research Dissemination Award, University of Waterloo
- 2020-2025 Vector Institute Research Grant
- 2019-2020 University of Manitoba Graduate Funding (UMGF)
- 2013-2017 People's Scholarship of China
- 2016 China Scholarship Council Scholarship for Studying Abroad
- 2016 UESTC Exchange Award

Professional Services

- 2026 **IT Professor Advisory Committee**, *University of Ottawa*.
- 2024-2025 **Equity, Diversity, and Inclusion (EDI) Committee**, *University of Waterloo*.
Graduate student representative.
- 2024 **School Advisory Committee on Appointments (SACA)**, *University of Waterloo*.
Graduate student representative and student host for faculty candidate meetings.
- 2024 **Graduate Recruiting Committee**, *University of Waterloo*.
Graduate student representative and an invited panelist for the grad student visiting.
- 2021-2025 **Conference Reviewer**, *NeurIPS (2022-2025)*, *ICML (2022-2025)*, *ICLR (2024-2025)*, *AAAI (2021-2022, 2025)*, *AISTATS (2025)*.
- 2021-2025 **Journal Reviewer**, *Transactions on Machine Learning Research*, *Neural Networks*, *IEEE Transactions on Multimedia*, *IEEE Transactions on Circuits and Systems for Video Technology*.

Publications

Journal Papers

- [1] **MUC: Machine Unlearning for Contrastive Learning with Black-box Evaluation**, Yihan Wang*, **Yiwei Lu***, Guojun Zhang, Franziska Boenisch, Adam Dziedzic, Yaoliang Yu, Xiao-Shan Gao, in *Transactions on Machine Learning Research (TMLR)*, 2025 (also appeared in *ICML 2024 Next Generation of AI Safety Workshop (Oral)*).
- [2] **f-MICL: Understanding and Generalizing InfoNCE-based Contrastive Learning**, **Yiwei Lu***, Guojun Zhang*, Sun Sun, Hongyu Guo, Yaoliang Yu, in *Transactions on Machine Learning Research (TMLR)*, 2023 (also appeared in *NeurIPS 2021 Workshop on Self-supervised Learning (Contributed Talk)*).
- [3] **Indiscriminate Data Poisoning Attacks on Neural Networks**, **Yiwei Lu**, Gautam Kamath, Yaoliang Yu, in *Transactions on Machine Learning Research (TMLR)*, 2022 (also appeared in *NeurIPS 2022 ML Safety Workshop and Trustworthy and Socially Responsible Machine Learning (TSRML) Workshop*).
- [4] **AdaCrowd: Unlabeled Scene Adaptation for Crowd Counting**, Mahesh Kumar Krishna Reddy, Mrigank Rochan, **Yiwei Lu**, Yang Wang, in *IEEE Transactions on Multimedia*, Volume 24, Page 1008-1019, 2022.
- [5] **Structure Learning with Similarity Preserving**, Zhao Kang, Xiao Lu, **Yiwei Lu**, Chong Peng, Wenyu Chen, and Zenglin Xu, in *Neural Networks*, 2020, Volume 129, Page 138-148.

Conference Papers

- [6] **Demystifying Foreground-Background Memorization in Diffusion Models**, Jimmy Z Di*, **Yiwei Lu***, Yaoliang Yu, Gautam Kamath, Adam Dziedzic, Franziska Boenisch, in *AAAI 2026*.
- [7] **BridgePure: Revealing the Fragility of Black-box Data Protection**, Yihan Wang*, **Yiwei Lu***, Xiao-Shan Gao, Yaoliang Yu, Gautam Kamath, in *NeurIPS 2025*.
- [8] **Machine Unlearning Fails to Remove Data Poisoning Attacks**, Martin Pawelczyk, Jimmy Z. Di, **Yiwei Lu**, Gautam Kamath, Ayush Sekhari, Seth Neel, in *ICLR 2025* (also appeared in *ICML 2024 Generative AI and Law Workshop (Spotlight)*).
- [9] **Disguised Copyright Infringement of Latent Diffusion Models**, **Yiwei Lu***, Matthew Y.R Yang*, Zuoqiu Liu*, Gautam Kamath, Yaoliang Yu, in *ICML 2024*.
- [10] **Indiscriminate Data Poisoning Attacks on Pre-trained Feature Extractors**, **Yiwei Lu**, Matthew Y.R Yang, Gautam Kamath, Yaoliang Yu, in *IEEE SaTML 2024*.
- [11] **Understanding Neural Network Binarization with Forward and Backward Proximal Quantizers**, **Yiwei Lu**, Yaoliang Yu, Xinlin Li, Vahid Partovi Nia, in *NeurIPS 2023*.

- [12] **Exploring the Limits of Model-Targeted Indiscriminate Data Poisoning Attacks**, **Yiwei Lu**, Gautam Kamath, Yaoliang Yu, in *ICML 2023*.
- [13] **Few-shot Scene-Adaptive Anomaly Detection (Spotlight)**, **Yiwei Lu**, Frank Yu, Mahesh Kumar K and Yang Wang, in *ECCV 2020*.
- [14] **Future Frame Prediction Using Convolutional VRNN for Anomaly Detection**, **Yiwei Lu**, Mahesh Kumar K, Seyed shahabeddin Nabavi and Yang Wang., in *IEEE AVSS 2019*.
- [15] **Similarity Learning via Kernel Preserving Embedding**, Zhao Kang, **Yiwei Lu**, Yuanzhang Su, Changsheng Li, Zenglin Xu, in *AAAI 2019*.
- [16] **Homoglyph Attack Detection with Unpaired Data**, **Yiwei Lu**, Mahesh Kumar K, Noman Mohammed, Yang Wang, in *ACM/IEEE Symposium on Edge Computing 2019*.

Workshop Papers

- [17] **On the Robustness of Neural Networks Quantization against Data Poisoning Attacks**, **Yiwei Lu**, Yihan Wang, Guojun Zhang, Yaoliang Yu, in *ICML 2024 Next Generation of AI Safety Workshop*.
- [18] **CM-GAN: Stabilizing GAN Training with Consistency Models**, Haoye Lu, **Yiwei Lu**, Dihong Jiang, Spencer Ryan Szabados, Sun Sun, Yaoliang Yu, in *ICML 2023 Workshop on Structured Probabilistic Inference & Generative Modeling*.
- [19] **Semantic Segmentation in Compressed Videos**, Ang Li*, **Yiwei Lu***, Yang Wang, in *IEEE International Workshop on Multimedia Signal Processing 2019*.

Preprint

- [20] **Not All Samples Are Equal: Quantifying Instance-level Difficulty in Targeted Data Poisoning**, William Xu*, **Yiwei Lu***, Yihan Wang, Matthew YR Yang, Zuoqiu Liu, Gautam Kamath, Yaoliang Yu, *Preprint (2025)*.

Talks (Selected)

- 2025 **Vector Machine Learning Security and Privacy Workshop**, *Adversarial Perturbation for Data Protection: Limitations and Pitfalls*.
- 2025 **University of Ottawa**, *Trustworthy Machine Learning with Data in the Wild*.
- 2025 **Concordia University**, *Trustworthy Machine Learning with Data in the Wild*.
- 2025 **University of Texas at Arlington**, *Trustworthy Machine Learning with Data in the Wild*.
- 2025 **University of Victoria**, *Trustworthy Machine Learning with Data in the Wild*.
- 2025 **Western University**, *Trustworthy Machine Learning with Data in the Wild*.
- 2024 **University of Waterloo (ECE)**, *Copyright Issues in Generative Models and Countermeasures*.
- 2024 **UESTC**, *Trustworthy Machine Learning with Data in the Wild*.
- 2024 **Vector Machine Learning Security and Privacy Workshop**, *Disguised Copyright Infringement of Latent Diffusion Models*.
- 2024 **IEEE SaTML**, *Indiscriminate Data Poisoning Attacks on Pre-trained Feature Extractors*.
- 2023 **Fudan University**, *Exploring the Limit of Indiscriminate Data Poisoning Attacks*.
- 2022 **National Research Council Canada**, *f-Mutual Information Contrastive Learning*.

Teaching

- Winter 2026 **Instructor**, *University of Ottawa*.
 SED3115: Introduction to Artificial Intelligence
 SYS5185/IAI5101/GNG5300: Foundations and Applications of Machine Learning
- 2024 **Open Course Development**, *University of Waterloo & Cybersecurity and Privacy Institute*.
 Artificial Intelligence (AI) Literacy: module on "Data Poisoning Attacks and Defenses".

- 2020-2024 **Teaching Assistant**, *University of Waterloo*.
CS480/680: Introduction to Machine Learning
CS245: Logic and Computation
CS116: Introduction to Computer Science 2
- 2018-2020 **Teaching Assistant**, *University of Manitoba*.
COMP4550: Real-time Systems
COMP4190: Advanced Artificial Intelligence

Work Experiences

- 2022 **Research Internship**, *Huawei Noah's Ark Lab, Montreal*.
Machine Learning Researcher on Neural Network Quantization
Published a paper in NeurIPS 2023.
- 2021 **Research Internship**, *National Research Council Canada, Waterloo*.
Machine Learning Researcher on Contrastive Learning
Published a paper in NeurIPS 2021 Workshop and TMLR 2023.

Supervision of Research Students

- 2026-present **Shivani Singal**, *University of Ottawa*, NSERC USRA.
- 2026-present **Christian Paravalos**, *University of Ottawa*, Undergraduate Researcher.
- 2026-present **Chenyu Zhang**, *University of Waterloo*, Undergraduate Researcher.
- 2025-present **Jack Anderson**, *University of Chicago*, Vector Internship.
- 2024-2025 **William Xu**, *University of Waterloo*, Undergraduate Researcher, Now software engineer at Waabi.
- 2023-2024 **Robert Liu**, *University of Waterloo*, Undergraduate Researcher, Now machine learning engineer at Google.
- 2023-2024 **Matthew Y.R Yang**, *University of Waterloo*, Undergraduate Researcher, Now PhD student at MIT.
- 2019-2020 **Frank Yu**, *University of Manitoba*, Undergraduate Researcher, Now research engineer at Meta.
- 2019-2020 **Ang Li**, *University of Manitoba*, Undergraduate Researcher, Now machine learning engineer at Primate Labs Inc.